

Serverless Image Processing Pipeline

Project: Serverless Image/Video Processing

Services: Amazon S3, AWS Lambda, AWS Step Functions, Amazon Rekognition, Amazon SNS

Description: Automate image processing workflows using AWS serverless services. When an image is uploaded to Amazon S3, AWS Lambda triggers an AWS Step Functions workflow to resize the image, detect labels using Amazon Rekognition, and send notifications through Amazon SNS.

Skills: Serverless Computing, Event-Driven Architecture, Workflow Automation, Image Processing, AWS AI Services, Lambda Scripting.

Definition: The Serverless Image Processing Pipeline is designed to automatically process uploaded images using AWS cloud-native services. The system uses Amazon S3 for image storage, AWS Lambda for serverless execution, AWS Step Functions for workflow orchestration, Amazon Rekognition for AI-based label detection, and Amazon SNS for sending notifications. Whenever an image is uploaded to S3, the workflow automatically resizes the image, analyzes image labels, and sends a processing result notification to the user.

Scope of the Project: The scope of this project is to design and implement a fully automated serverless image processing pipeline using AWS services. The project leverages event-driven architecture and serverless orchestration to process uploaded images automatically without manual intervention.

This project covers:

- **Automated Image Upload Processing:** Automatically triggers processing workflows when an image is uploaded to Amazon S3.
- **Image Resizing:** Automatically resizes uploaded images and stores resized images into another S3 bucket.
- **AI-Based Label Detection:** Uses Amazon Rekognition to detect labels and objects from uploaded images.
- **Automated Notifications:** Sends processing results and detected labels through Amazon SNS email notifications.
- **Serverless Workflow Automation:** Uses AWS Step Functions to orchestrate multiple Lambda functions within a single automated workflow.

Methodology:

1. Lambda + Step Functions + Rekognition – Best Method

- Uses all required AWS services: S3, Lambda, Step Functions, Rekognition, and SNS.
- Provides fully automated image processing workflows using serverless architecture.
- Automatically resizes images, detects labels, and sends notifications without manual intervention.
- AWS Step Functions provide centralized orchestration and monitoring of workflow execution.
- Event-driven architecture ensures real-time processing of uploaded images.
- Cost-effective and free-tier friendly for small workloads.
- Production-ready architecture suitable for scalable cloud deployments.

2.EC2 + Python Image Processing – Alternative Method

- Uses EC2 instances with Python image processing libraries and custom scripts.
- Allows deep customization of image processing logic and workflow management.
- Requires manual server management, scaling, patching, and monitoring.
- Involves higher operational cost and infrastructure maintenance compared to serverless architecture.
- Less efficient for dynamic scaling and event-driven processing.
- Suitable for legacy workloads or applications requiring persistent compute resources.

AWS Services used in the Project:

1.Amazon S3 (Simple Storage Service)

- **Definition:** Amazon S3 is an object storage service that provides scalable, secure, and highly durable cloud storage.
- **Purpose:** Stores uploaded images and resized output images.
- **Role:** Acts as the source bucket for image uploads and destination bucket for resized image storage.

2.AWS Lambda

- **Definition:** AWS Lambda is a serverless compute service that allows users to run code without provisioning or managing servers.
- **Purpose:** Executes serverless image processing tasks automatically.
- **Role:** Triggers workflows, resizes images, detects labels, and sends notifications.

3.AWS Step Functions

- **Definition:** AWS Step Functions is a serverless workflow orchestration service used to coordinate multiple AWS services into automated workflows.
- **Purpose:** Controls and orchestrates the image processing workflow.
- **Role:** Coordinates image resizing, label detection, and SNS notification tasks sequentially.

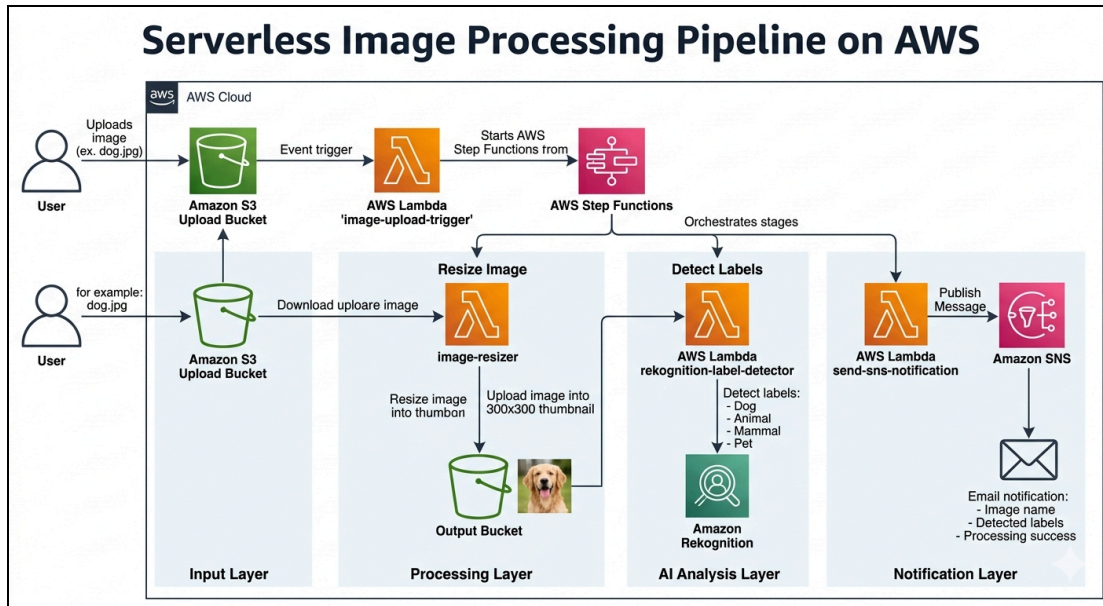
4.Amazon Rekognition

- **Definition:** Amazon Rekognition is an AWS AI service that provides image and video analysis using machine learning.
- **Purpose:** Detects labels, objects, and image metadata automatically.
- **Role:** Analyzes uploaded images and identifies labels such as animals, pets, vehicles, objects, etc.

5.Amazon SNS (Simple Notification Service)

- **Definition:** Amazon SNS is a fully managed messaging and notification service used for sending alerts and notifications.
- **Purpose:** Sends email notifications after successful image processing.
- **Role:** Delivers processing status and detected image labels to subscribed email users.

Architecture Diagram:

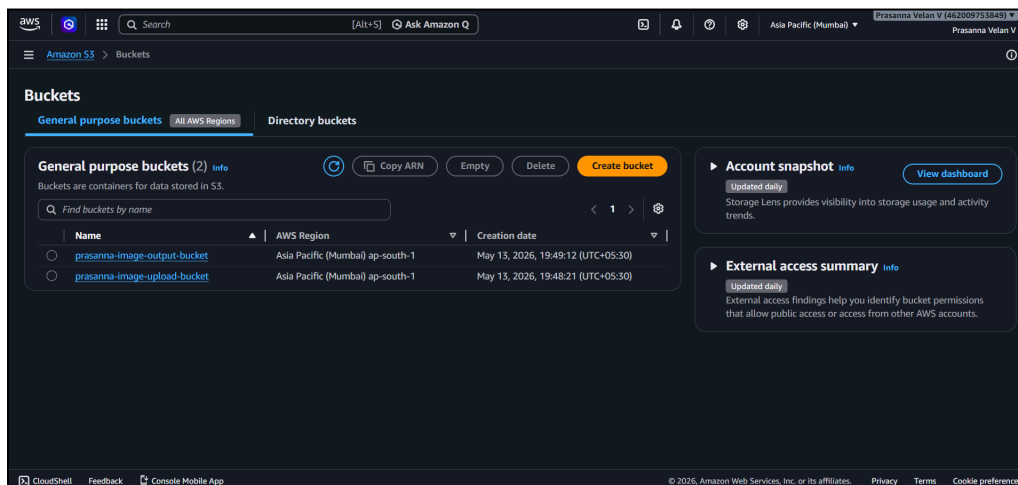


Implementation Steps to Do the Project:

Serverless Image Processing Pipeline

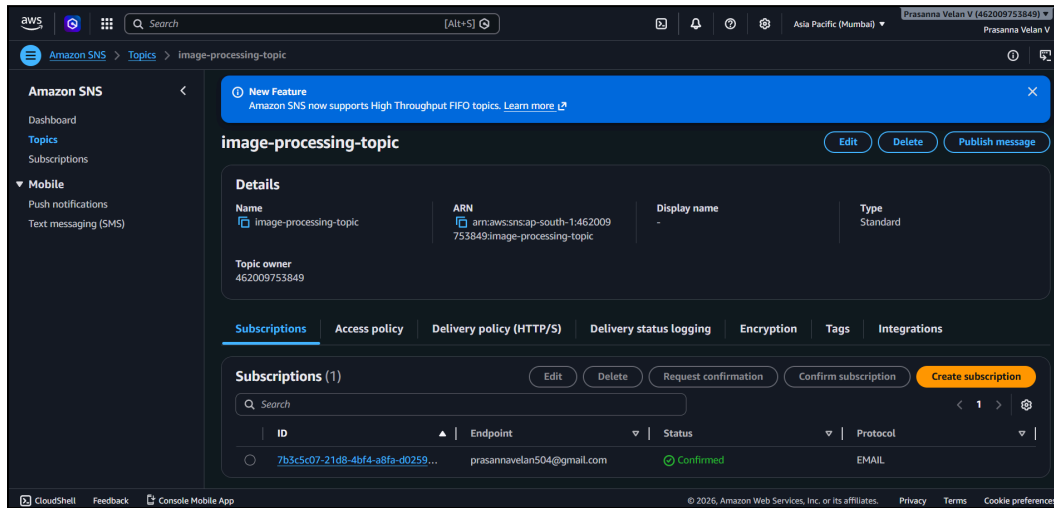
Step 1: Create S3 Buckets

- Go to AWS Console → S3 → Create Bucket.
- Created Upload Bucket: **prasanna-image-upload-bucket**
- Created Output Bucket: **prasanna-image-output-bucket**
- Create both buckets successfully.



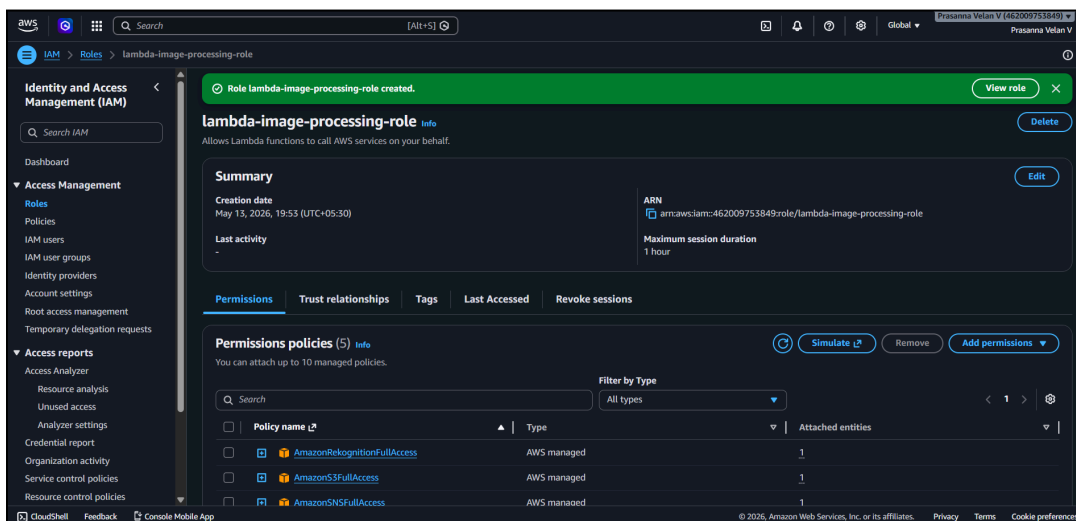
Step 2: Create SNS Topic

- Go to AWS Console → SNS → Topics → Create Topic.
- Selected Topic Type: Standard
- Topic Name: **image-processing-topic**
- Create Email Subscription.



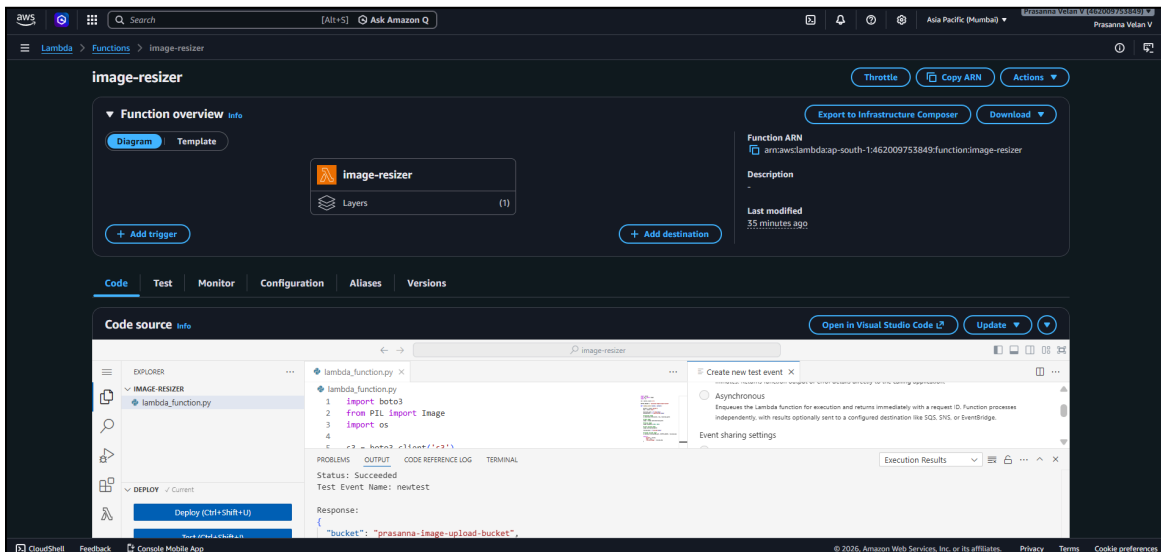
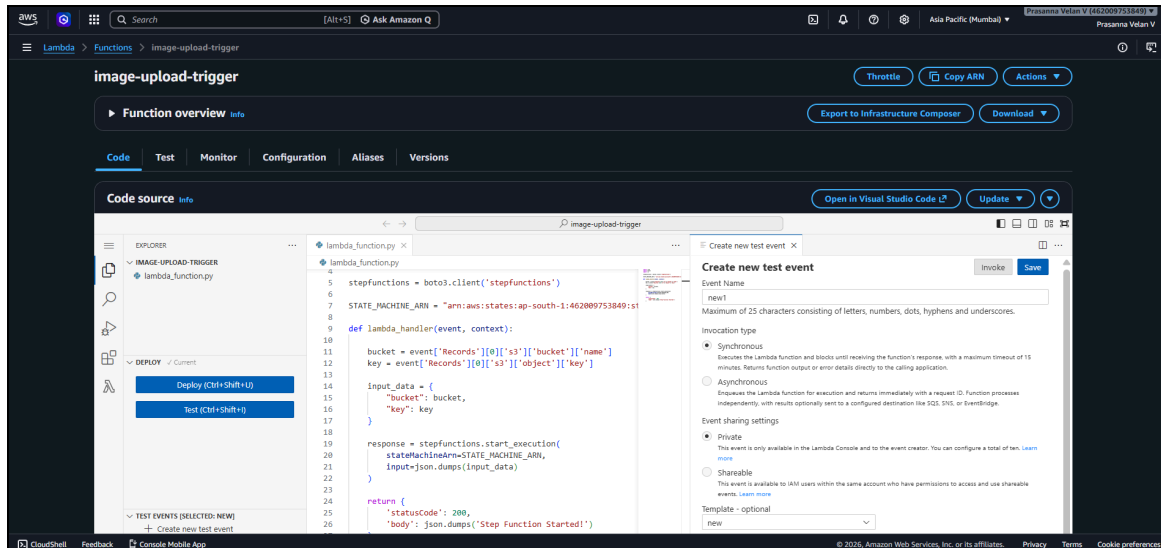
Step 3: Create IAM Role for Lambda

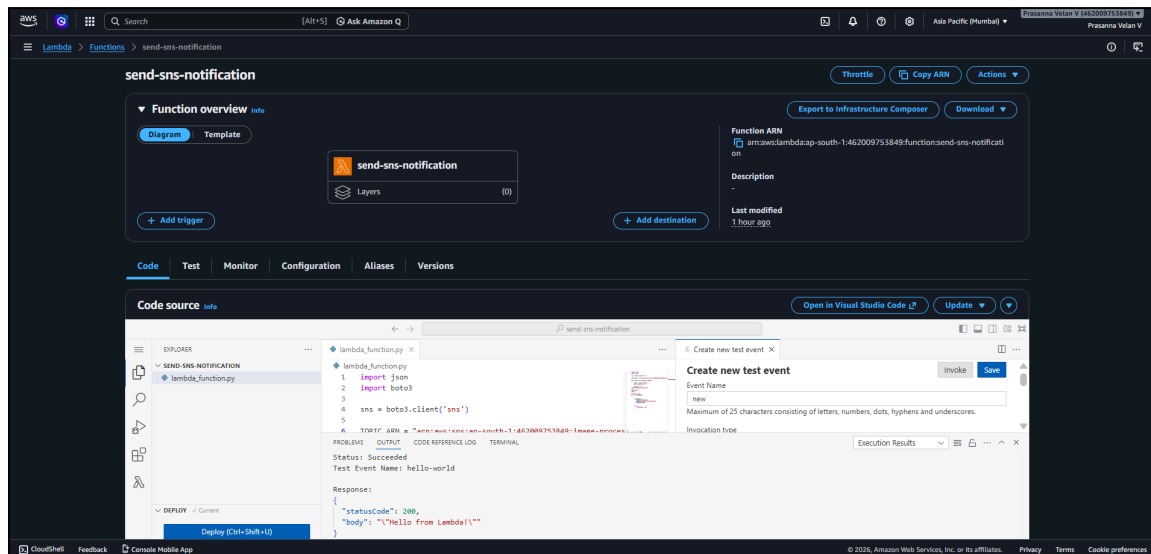
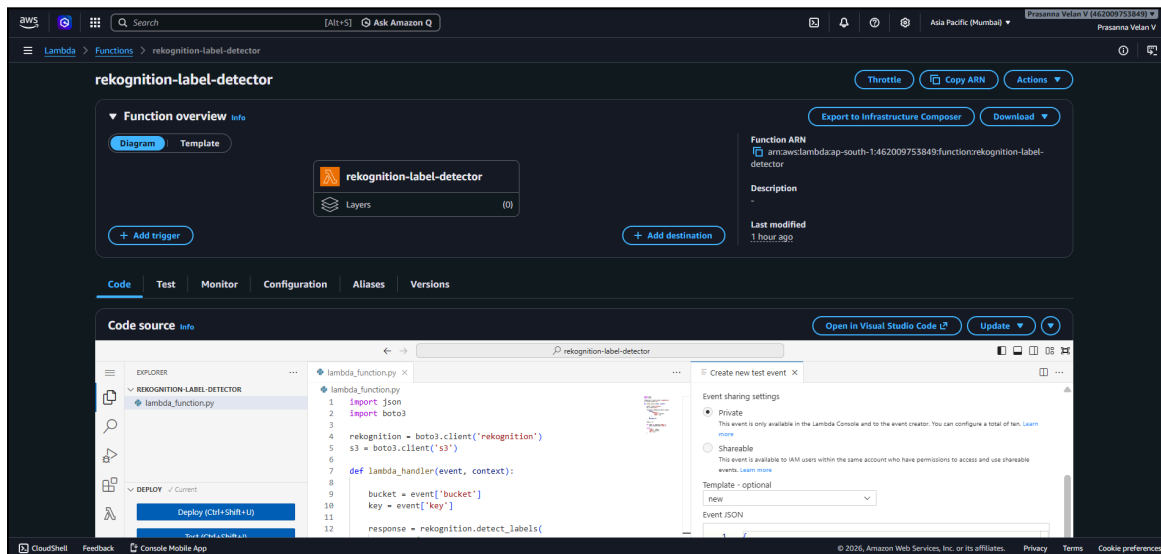
- Go to IAM → Roles → Create Role → Lambda.
- Attach Policies: AmazonS3FullAccess, AmazonRekognitionFullAccess, AWSStepFunctionsFullAccess, AmazonSNSFullAccess, CloudWatchLogsFullAccess
- Role Name: **lambda-image-processing-role**



Step 4: Create Lambda Functions

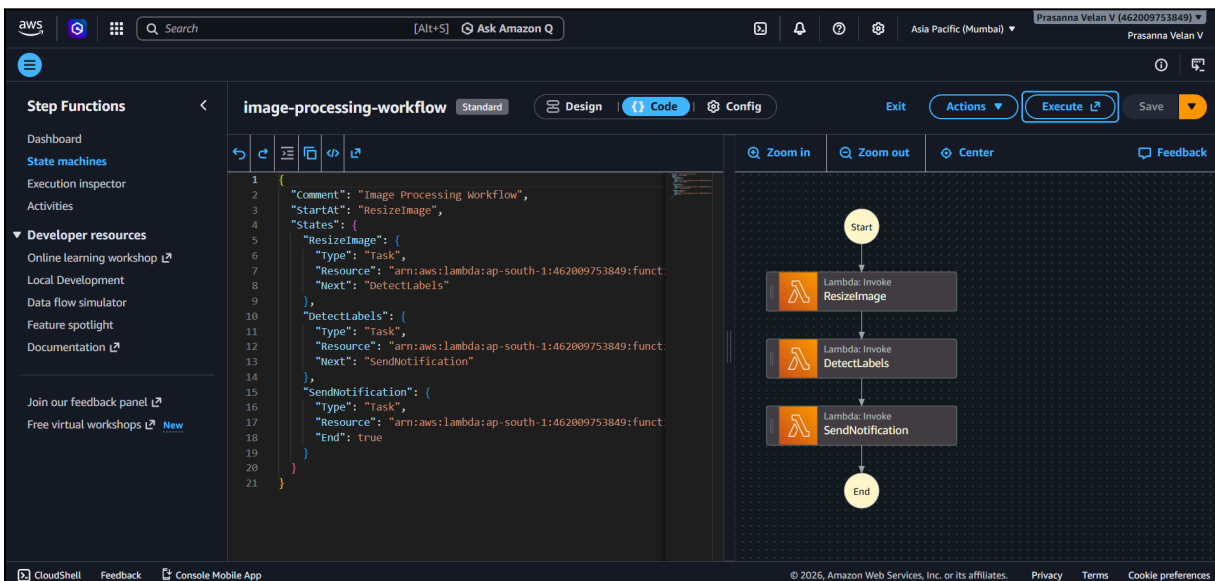
- Lambda → Create Function → Author from Scratch. Create the following Lambda functions:
- image-upload-trigger
- Triggered by S3 upload events and Starts AWS Step Functions workflow.
- **Image-resizer** : Downloads uploaded images from S3 and Resizes images into thumbnails. Uploads resized images into output bucket.
- **Rekognition-label-detector** : Uses Amazon Rekognition to detect labels from uploaded images.
- **Send-sns-notification** : Sends email notifications using Amazon SNS.
- Add Existing Role : **lambda-image-processing-role**





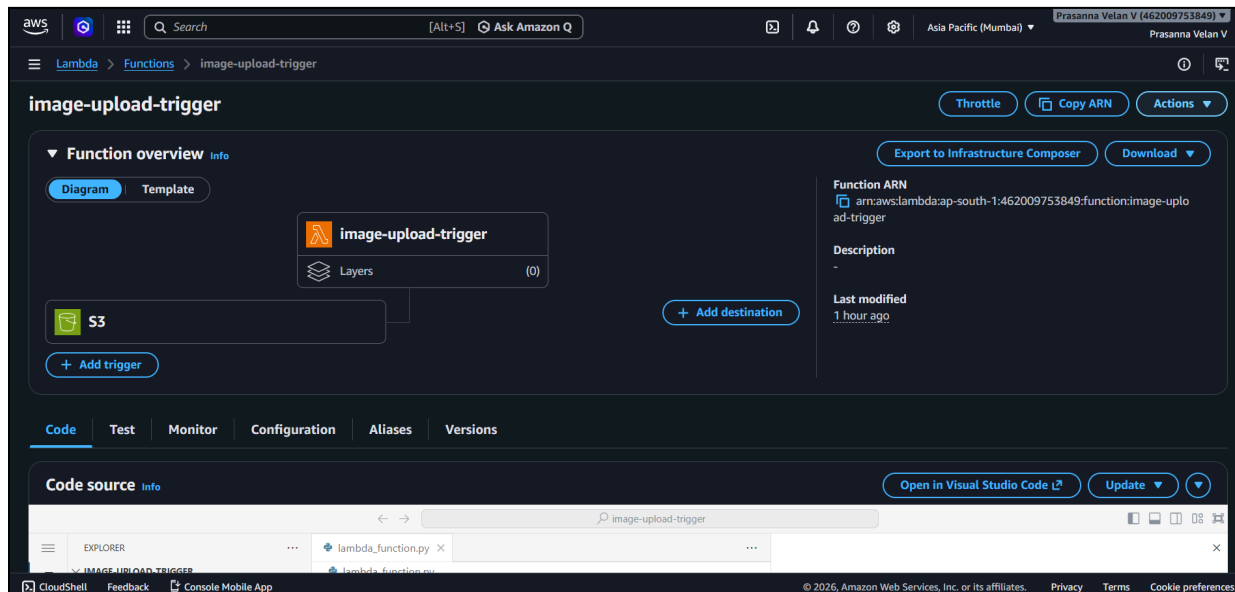
Step 5: Create Step Functions Workflow

- Go to AWS Step Functions → Create State Machine.
- Choose: Write Workflow in Code.
- Create workflow stages: **Resize Image ,Detect Labels ,Send Notification**
- Configure Lambda ARNs in workflow JSON and Save and deploy the workflow.



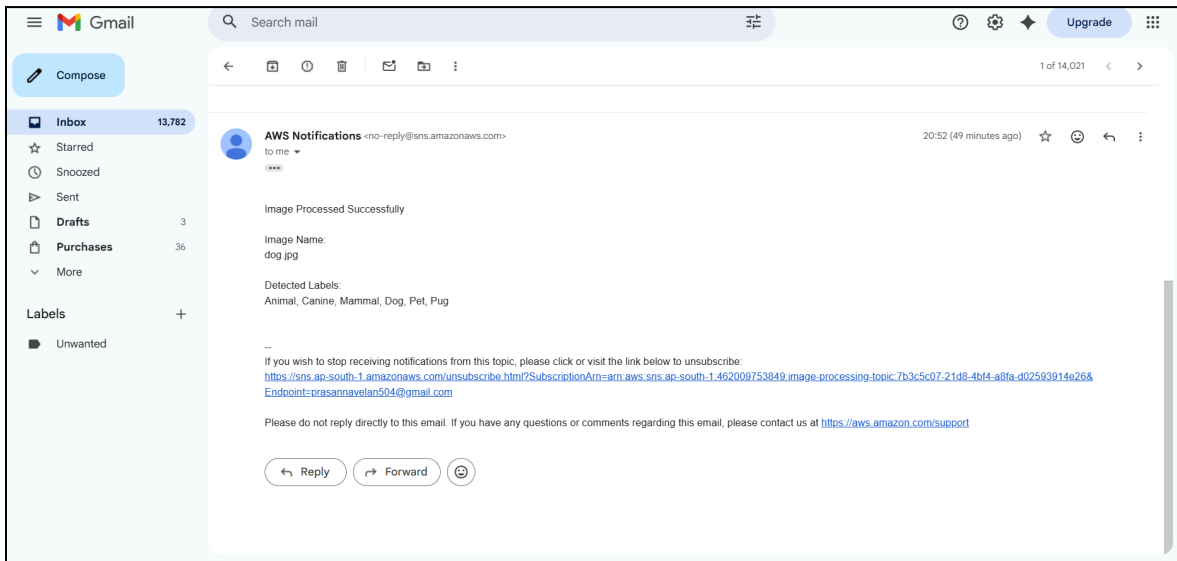
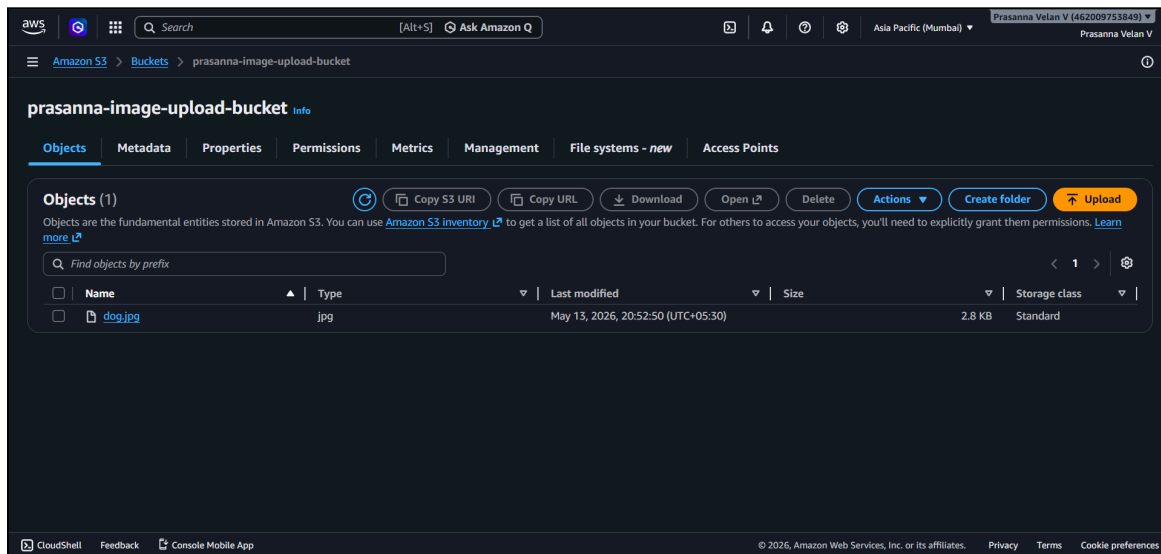
Step 6: Configure S3 Trigger

- Open Lambda Function: **image-upload-trigger**
- Add Trigger: **Amazon S3** , Event Type: **PUT / Object Created Events**
- Select Upload Bucket: **prasanna-image-upload-bucket**
- Enable trigger.



Step 7: Test the Project

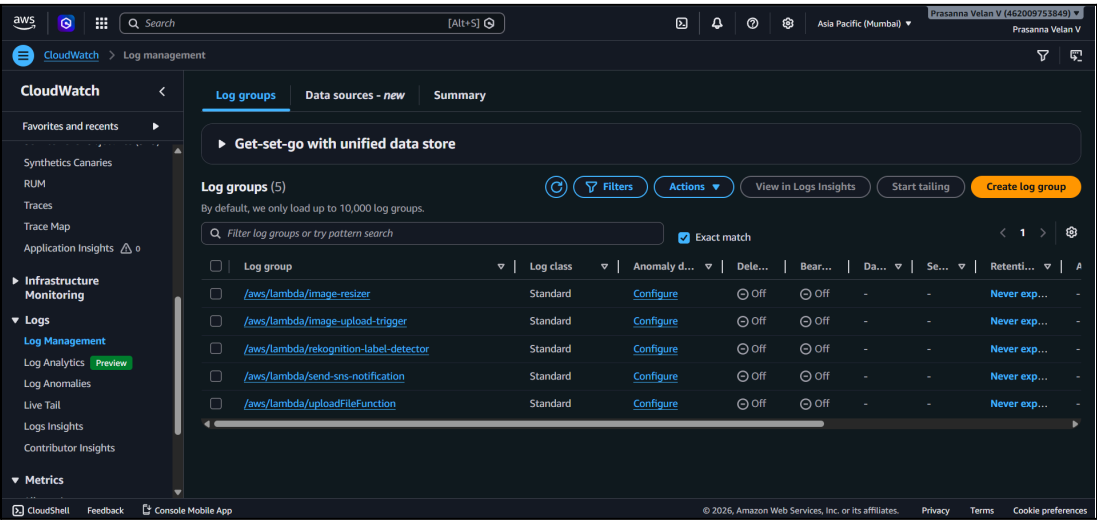
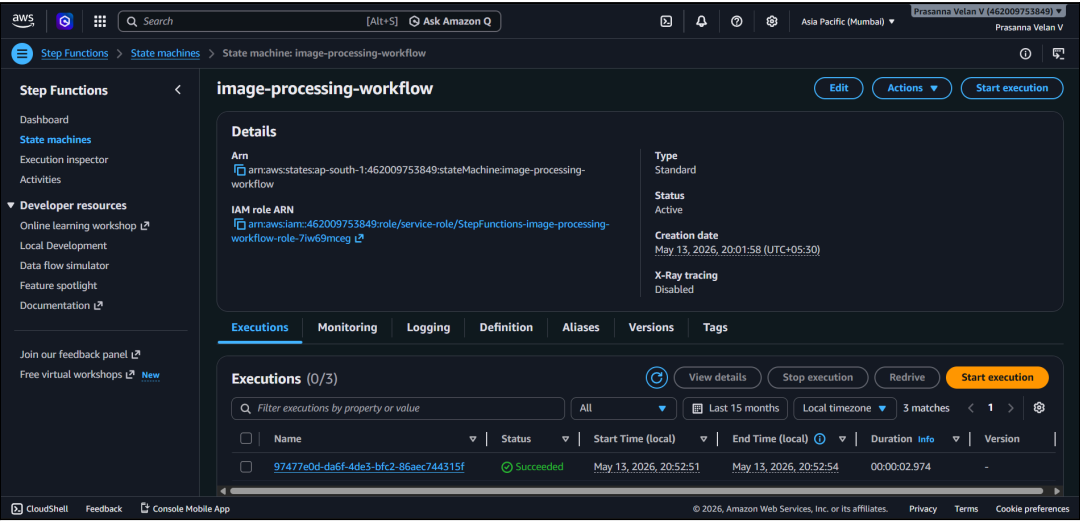
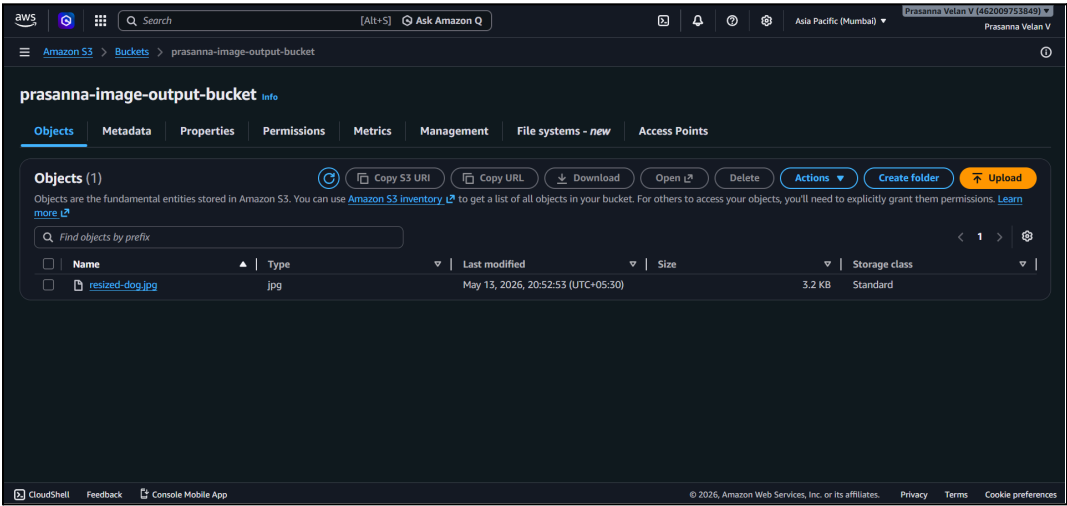
- Upload an image into the upload bucket. Example: **dog.jpg**
- Workflow execution automatically starts.
- Verify: **Image resized successfully.**
- Verify: **Labels detected using Rekognition.**
- Verify: **SNS email notification received successfully.**



Step 8: Verify Output and Workflow

- Go to S3 Output Bucket , **Confirm resized image availability.**
- Go to Step Functions → Executions ,Verify **successful workflow execution.**
- Go to CloudWatch Logs.
- Verify **Lambda execution logs and monitoring information.**

Serverless Image Processing Pipeline



Troubleshooting:

Issue 1: Lambda Layer Import Error

The image resizing Lambda failed due to missing Pillow image processing libraries.

Solution:

- Added AWS-compatible Pillow Lambda Layer.
- Attached compatible Pillow layer to image-resizer Lambda function.
- Redeployed the Lambda function.

Issue 2: Lambda Timeout Issue

Image processing tasks may exceed default Lambda timeout duration.

Solution:

- Go to Lambda → Configuration.
- Increase timeout duration appropriately.
- Save configuration changes.

Issue 3: SNS Notification Not Received

SNS email notifications failed because the subscription was not confirmed.

Solution:

- Open email inbox.
- Confirm SNS subscription.
- Retry image upload workflow.

Conclusion:

The Serverless Image Processing Pipeline provides a reliable, scalable, and fully automated solution for image processing using AWS cloud-native services. Using Amazon S3, AWS Lambda, AWS Step Functions, Amazon Rekognition, and Amazon SNS, the project automatically processes uploaded images by resizing them, detecting labels using AI services, and sending notifications without manual intervention.

The serverless architecture reduces operational overhead, supports event-driven automation, and provides scalable cloud processing capabilities. The project demonstrates workflow orchestration, AI integration, and automation using AWS services efficiently. It is suitable for organizational deployment, educational demonstrations, and cloud-native application development while remaining cost-effective and free-tier friendly for small workloads.